

The Cyber Challenge

- **Asymmetric Advantage:** Attackers need to find just one vulnerability; defenders must protect everything
- **Alert Fatigue:** 67% of alerts ignored due to volume
- **Base-Rate Fallacy:** Most alerts are false positives, causing real threats to be missed
- **Static Defenses Fail:** Predictable patterns exploited by attackers

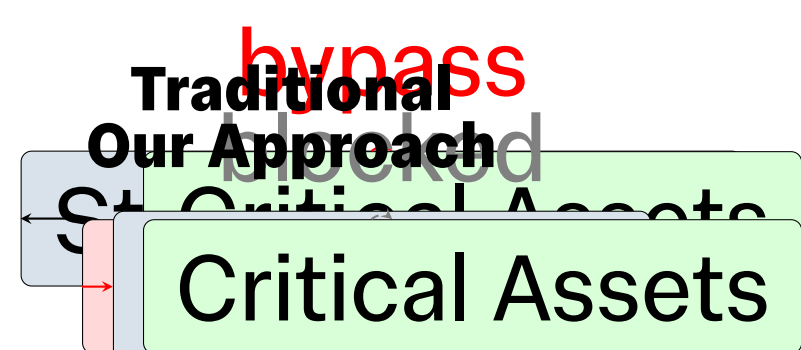


Figure 1: Static vs. adaptive RL approach.

"Traditional defenses remain static while attackers continuously adapt. An intelligent defense must adapt as quickly as the threat."

Research Aim

To develop and evaluate an RL Agent that learns optimal decoy placement strategies, maximizing attacker deception while minimizing defensive resource costs.

Research Questions:

1. Can RL effectively learn adaptive decoy deployment?
2. How does a learned approach compare to traditional methods?
3. What strategic behaviors emerge and how do they scale?

Method

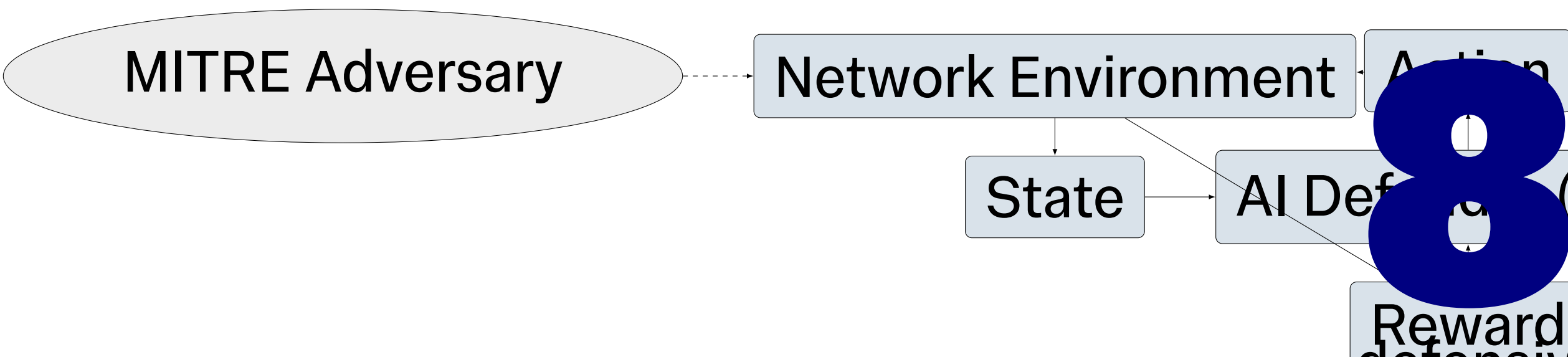


Figure 2: Reinforcement learning approach.

Environment:

- Multi-subnet enterprise network (15-200 hosts)
- Fixed attacker following MITRE ATT&CK tactics
- Decoys can be deployed or removed in any subnet

Reward Structure:

$$R_t = 10v_r\mathbf{1}[\text{decoy hit}] - v_r\mathbf{1}[\text{asset hit}] - 20\mathbf{1}[\text{deploy}] - \mathbf{1}[\text{remove}]$$

where v_r escalates with attack progression (recon → impact) and costs penalize excessive deployment

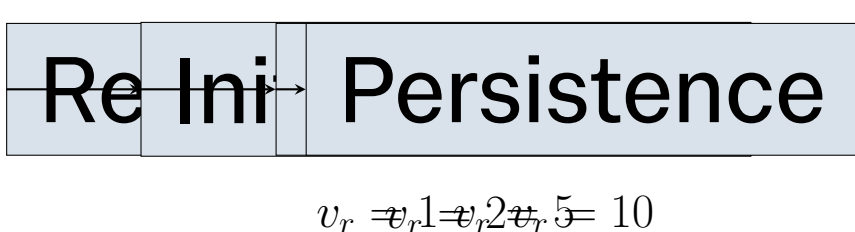


Figure 3: MITRE ATT&CK-based attack progression.

Comparison Baselines:

- **Static:** Fixed interval placement with capacity limits
- **Rule-based:** Temporal heuristics
- **Inactive:** No decoys (control)

Results

89% Improvement in defensive effectiveness

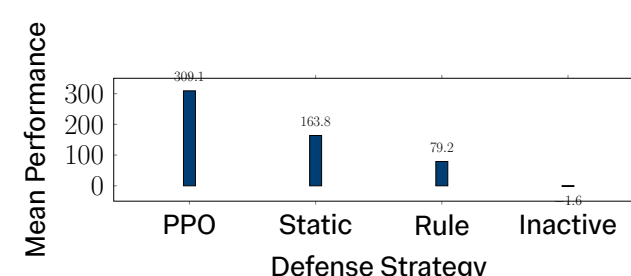


Figure 4: Performance comparison across strategies.

Metric	PPO	Static	Rule
Deception	73.2%	45.1%	32.4%
Protection	82.4%	61.7%	48.2%
Latency	2.3	4.1	5.8

Strategic Behavior

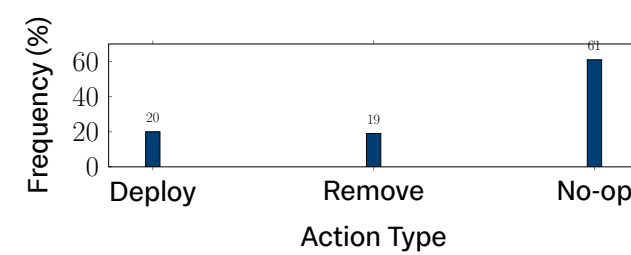


Figure 5: Action distribution showing strategic restraint.

Emergent Intelligence:

The AI learned that excessive decoy deployment is counterproductive - it increases costs without proportional security gains. This strategic restraint emerged naturally.

Network Visualization

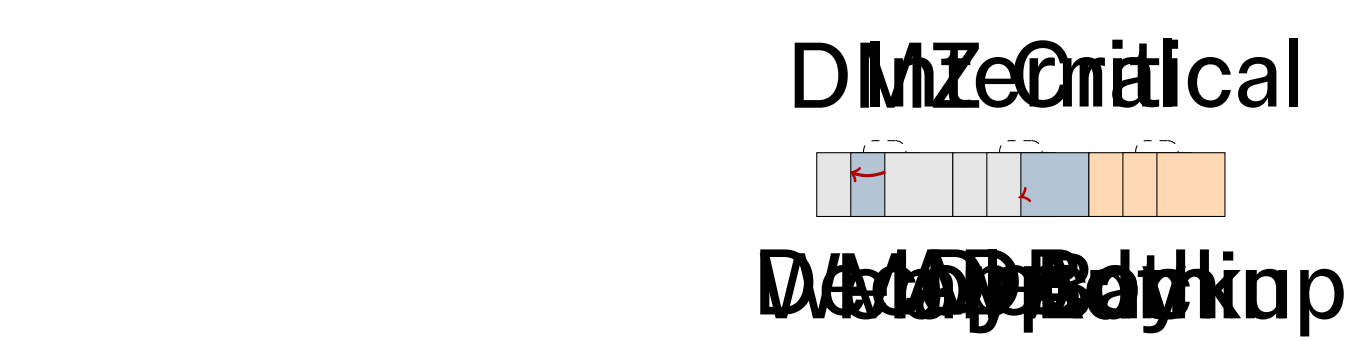


Figure 6: Network with AI-placed decoys intercepting attacks.

Scaling Performance

Table 2: Performance across network sizes.

Network	Reward	Retention
15 hosts	309.1	100%
50 hosts	325.4	105%
200 hosts	298.2	96%

Conclusion

Key Findings:

- AI successfully learns adaptive cyber defense strategies
- 89% improvement over traditional approaches
- Strategic restraint emerges naturally
- Performance scales to enterprise networks

Statistical validation: 5 seeds, 95% confidence intervals, ANOVA $p < 0.001$

Future Work

- Adversarial co-evolution (self-play)

- Multi-objective optimization
- Real-world testing with human red teams
- Transfer learning across network topologies

Contact & References

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References: Sommer & Paxson (Base-rate fallacy), Schulman et al. (PPO), MITRE ATT&CK Framework